

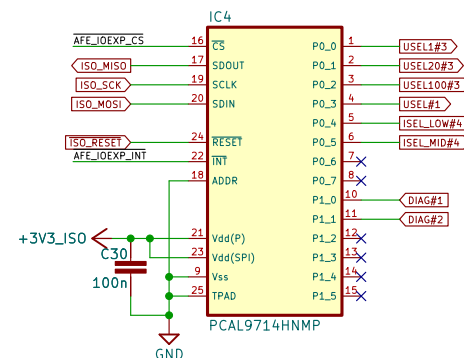
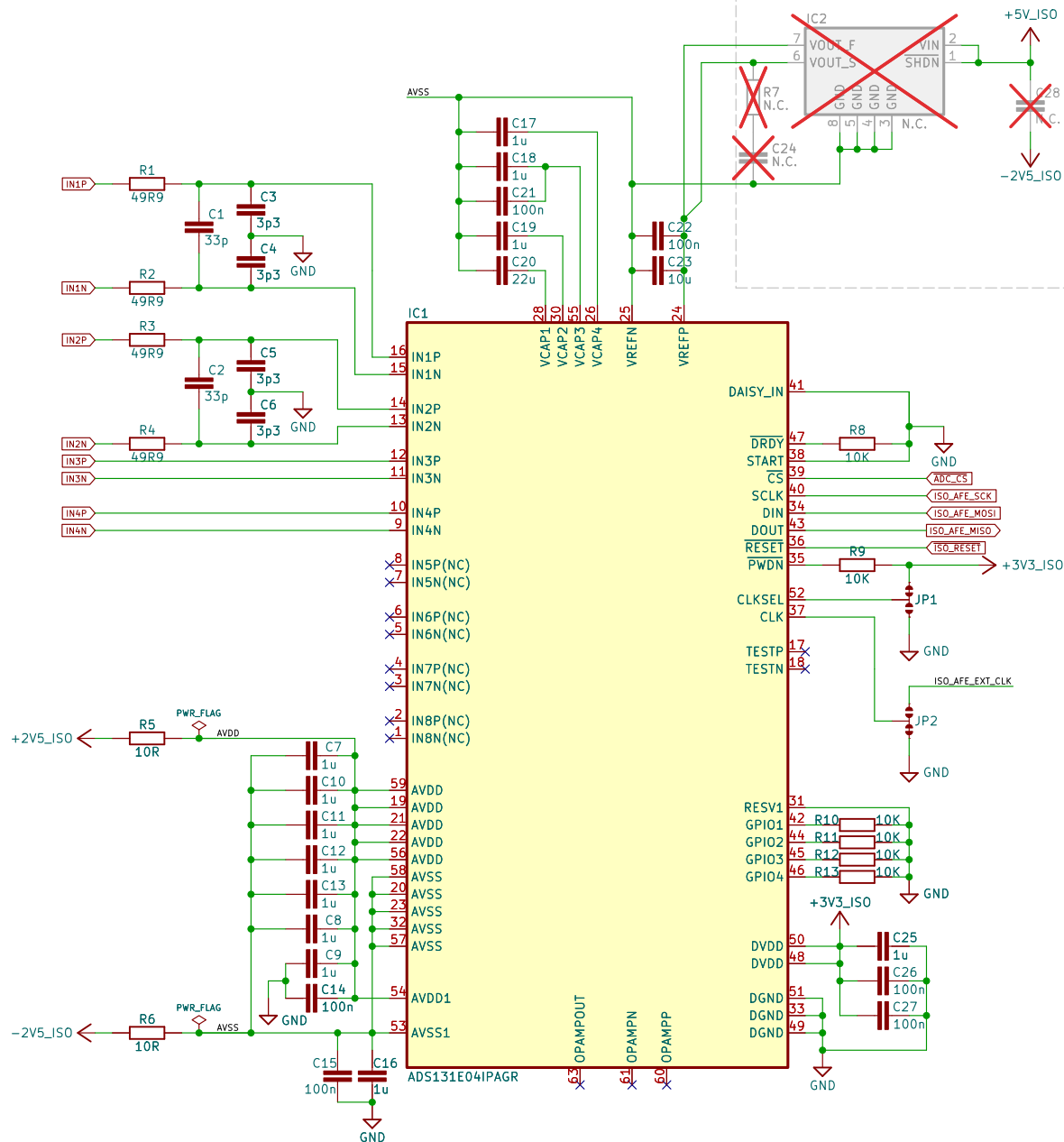
4-ch 24-bit ADC

+2.5 V voltage reference (optional)

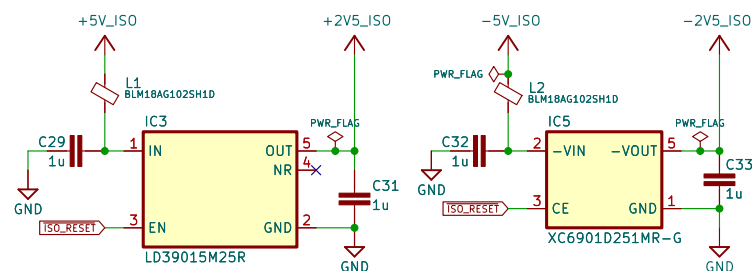
AFE connector

26-pin AFE connector

GND 1	2 GND
AFE_ID0 3	4 AFE_ID1
ISO_AFE_EXT_CLK 5	6 ISO_RESET
+3V3 7	8 +3V3
+5V 9	10 -5V
+15V 11	12 -15V
GND 13	14 GND
RELAY_CS 15	16 ADC_CS
SPI_SCK 17	18 AFE_CLK
SPI_MOSI 19	20 AFE_MOSI
SPI_MISO 21	22 AFE_MISO
AFE_IOEXP_CS 23	24 AFE_IOEXP_INT
GND 25	26 GND



+/-2.5 V LD0s



<https://www.envox.eu>

4-channel Isolated AFE for MIO168+

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Envox d.o.o.

Sheet: /ADC, IOexp, LD0s/

File: ADC.kicad_sch

Title: EEZ DIB AFE3+

Size: A4 Date: 2026-07-18

KiCad E.D.A. 10.0.4

Rev: r2B1

Id: 2/5

Isolated Voltage and Current amplifier

8-pin isolated two channels connector

CH#2 CS- 1
CH#2 CS+ 2
CH#2 Vin- 3
CH#2 Vin+ 4
CH#1 CS- 5
CH#1 CS+ 6
CH#1 Vin- 7
CH#1 Vin+ 8

X2
8 AIN1P
7 AIN1N
6 AIN2P
5 AIN2N
4 AIN3P
3 AIN3N
2 AIN4P
1 AIN4N

TBP02R1-381-08BE

F1 3561(KTR) x2
10A fuse

F2 3561(KTR) x2
1.25A fuse

K1B

K1C

K2C

K2B

R15 R200
low TCR ($\pm 25\text{ppm}/^\circ\text{C}$)

R16 R022
low TCR (max. $\pm 20\text{ppm}/^\circ\text{C}$)

L3

DLW215N3715Q2L

R19

10R

R20

10R

C36

8n2

L4

MLZ1608N100LT000

R18

470R

R21

1M5 0.1%

R22

1M5 0.1%

R23

1M5 0.1%

R24

82K5 0.1%

R17

10K5 0.1%

OK1

VOR1142B4

USEL#1

±1 V max.

C37

1u

C41

1n

C42

100n

C39

100n

C45

1n

C40

100n

C46

1n

IC7

AMC3330QDWER

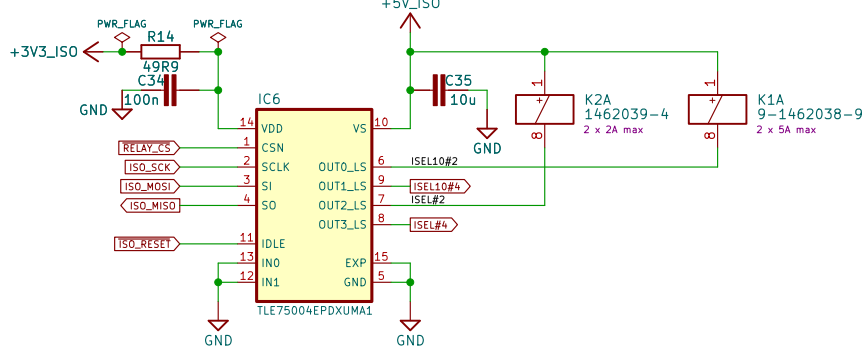
Fixed gain: x2
Voltage ranges: $\pm 50\text{ V}$, $\pm 480\text{ V}$

IC8

AMC3301DWER

Fixed gain: x8.2
Current ranges: $\pm 1\text{ A}$ (gain x1), $\pm 10\text{ A}$ (gain x2)

Relay driver



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4-channel Isolated AFE for MIO168+

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Sheet: /Isolated Ch#1, Ch#2/

File: Isolated_Ch#1_Ch#2.kicad_sch

Title: EEZ DIB AFE3+

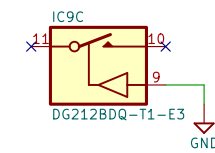
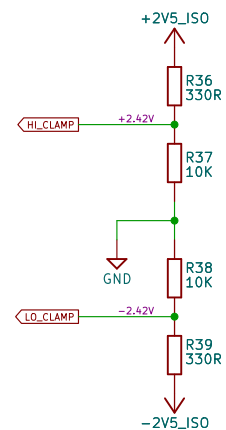
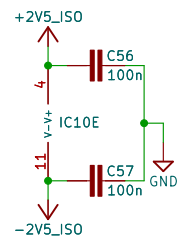
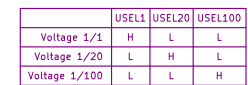
Size: A4

Date: 2026-07-18

KiCad E.D.A. 10.0.4

Rev: r2B1

Id: 3/5



Rev: r2B1
Id: 4/5

